GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
Client	Brass Fertilizer and Petrochemical Company Limited
Originating Company	Epic Atlantic Limited
Document Title	**HSSE Design Philosophy**
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HSSE Design Philosophy

R02	13.07.22	Re-issued for Review	E. Okoye	E. Maduabuchi	M. Ekpunobi	
R01	15.05.22	Issued for Review	E. Okoye	E. Maduabuchi	K. Yusuff	
Rev #	Issue Date	Status Description	Originator	Reviewer	Approver (Epic)	Approver (BFPCL)

Revision History

Rev No.	Date	Description of Revision
R01	15-05-22	Issued for Review
R02	13-07-22	Re-Issued for Review

Revision Philosophy:

a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.)
for the subsequent project phases, post FEED, it is expected that:
c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
e. All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

SUMMARY PAGE

Client:	Brass Fertilizer & Petrochemical Company Limited	
Project:	Gas Processing & Methanol Complex	**Project No.** 11998A
Facility:	Gas Gathering Pipelines & Associated Infrastructure	**Doc No.** EA-BFPCL-GPMC-GGPL-FEED-THS-PHY-002

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS: BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, MULTIPHASE FLOW, ROW, PTS

REFERENCES	
Document No.	Description
EA-BFPCL-GPMG-GGPL-FEED-PSE-RPT-001	Upstream Gas Gathering Concept Select & Revalidation Report
EA-BFPCL/BFPP GAS PL/FEED-PRO-RPT-001	Preliminary Simulation – Based on Desktop Route Survey
EA/SPDC-BFPC UGS-GEP/CSM-RPT-002	Gas Gathering & Evacuation Pipelines Concept Screening
EA/SPDC-BFPC UGS-GEP/CSM-RPT-001	Conceptual Study Memorandum
EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002	Basis of Design
	FISAS Draft ESIA Report (November 2020)

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Project Information, Terms, Abbreviations, and Definitions

Project Information – General

1.0	Employer	Brass Fertilizer & Petrochemical Company Limited (BFPCL)
2.0	PMC	Tata Consulting Engineers Limited (TCE)
3.0	EPCC Contractor	China Tianchen Engineering Corporation (TCC)
4.0	Project Title	Gas Processing and Methanol Complex (GPMC)
5.0	Location	Akabel, Brass Island: • 85km South West of Port Harcourt, • 70km South of Yenagoa and • 90km West of Bonny Island.
5.1	State	Bayelsa State, Nigeria
5.2	Nearest Air Port, Sea Port. Railway Station	Port Harcourt (PHC), 85km Bayelsa Airport (Under Construction)
6.0	Standard Process Condition	15°c and 1.01325 bar
7.0	Design Life	30 years

Definition of Terms

Terms	Definitions
BFPCL	Brass Fertilizer & Petrochemical Limited.
SPDC	Shell Petroleum Development Company Nig. Limited
Employer	Party (Usually BFPCL) that initiates the project and ultimately pays for its design and construction. The Employer will generally specify the technical requirements. The Employer may also include an agent or consultant authorized to act for and on behalf of, the Employer.
Engineer	The party that carries out the FEED Engineering of the project.
Manufacturer/Supplier	Party, which manufactures or supplies equipment and services to perform the duties specified by the Contractor.
Shall	The word 'shall' indicate a mandatory requirement.
Should	The word 'should' indicate a recommendation.
Measuring Range	Range defined by LRL and URL within which the limits of uncertainty of the measuring instrument are specified (the capability of the instrument).
On-Line Instruments	Instruments connected to process and utility lines or equipment via primary isolation valves.

Project	Front End Engineering Design for Gas Gathering Pipelines (Phase1)

Abbreviations

Abbreviation	Meaning
ALARP	As Low As Reasonably Practicable
DB&B	Double Block and Bleed
EGASPIN	Environmental Guidelines and Standards for Petroleum Industry in Nigeria
EIA	Environmental Impact Assessment
ESD	Emergency Shutdown
FEED	Front End Engineering Design
FRED	Fire Radiation Emission and Dispersion
GHG	Green House Gas
F&G	Fire and Gas
HAZID	Hazard Identification
HAZOP	Hazard Operability Studies
HEMP	Hazard & Effect Management Process
HFE	Human Factor Engineering
HSSE	Healthy Safety Environment and Security
MAH	Major Accident Hazard
PEFS	Process Engineering Flow Scheme
PFP	Passive Fire Protection
SCE	Safety Critical Element/ Equipment
SP	Social Performance

Site and Environmental Conditions*

1.	Air Temperature	Average daytime 32degC; rainy season 28degC; night time 23degC
2.	Weather	Dry Season (Harmattan); November – January; the atmosphere is laden with extremely fine dust (5 microns particle size) Wet Season: heavy rainfall, two peaks, August- 245mm, September- 331mm
3.	Relative Humidity	Mean 81%; Minimum 69% & Maximum 88%
4.	Wind Speed and Direction	Monthly mean 8.5km/h; varies from 6.2km/h to 9.6km/h The south-westerly wind is 60% predominant during the wet season (March - October). The north-easterly wind is 32% predominant during the dry season (November– February)
5.	Seismicity	The project area is classified as Seismic Zone 0

*Data from ESIA Report

1. Introduction

1.1. Background Information

Brass Fertilizer and Petrochemical Company Limited (BFPCL) is developing a 10,000 metric tonnes per day (MTPD) capacity greenfield methanol production facility in Akabel (Odioma), Brass LGA, Bayelsa state. The plant complex is to be based on natural gas feedstock from 5 SPDC fields in OMLs 33, 35 & 36.

The Project involves the development, construction, and operation of a plant comprising:

- Two trains of 5,000 MTPD methanol plant, producing a total of 10,000 MTPD of methanol
- 340 MMscfd Gas Processing Plant ("GPP"), which will extract NGL from the natural gas feedstock, prior to feeding the lean gas to the methanol plant.
- Gas gathering pipelines and associated manifolds transporting the natural gas from five fields, viz: Bubouwe Bou (BBOU), Elepa (ELEP), Nun River (NUNR), Seibou (SEIB), and Opugbene (OPUG) to the GPP at Akabel.
- Product Export Lines and SBM for the export of the methanol, and NGL of the GPP.
- All offsite, utility, and off-plot facilities (including product storage, internal power generation, steam generation, and export jetty) for the operation of the processing plants.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2.

The focus of the present FEED is on Phase 1 as shown in figure 1 below, while Phase 2 will be implemented later.

TCE is the Project Management Consultant for the project.

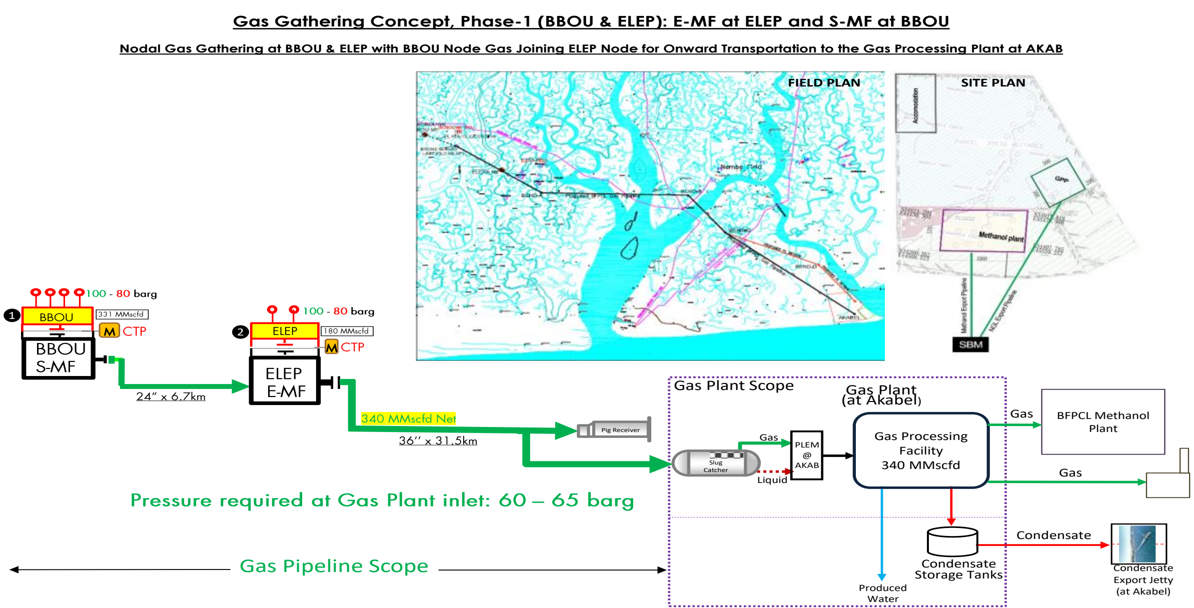

Figure 1: Gas Gathering Concept, Phase 1

1.2. Purpose

The objective of this document is to establish the Health, Safety, Security and Environmental (HSSE) principles that will guide the design of the surface facilities for the Project, to ensure that residual HSSE risk is as low as reasonably practicable (ALARP). This is a live document that will be updated as the project progresses.

The fundamental HSSE objective underlying the Project is that, on handover of the facility to Operations, residual HSSE risks are both tolerable and ALARP. This philosophy provides a road map for achievement of ALARP with respect to fire, gas and smoke release, and explosion protection in design.

Implicit in the achievement of ALARP is that systems (control and hardware) are engineered such that:

- Safety/HSSE criteria are achieved
- Systems are:
  - Fit for purpose, cost effective and function as intended;
  - Utilise the latest mature technology;
  - Are reliable and offer a high availability, are simple to operate and maintain (locally) and meet standardisation criteria;
  - Integrate with interfacing control and communications infrastructure;
  - Are flexible and accommodate future modifications, updates, and expansions;
  - Support the Project design life of a minimum of 25 years.

1.3. Scope

The scope of this philosophy document encompasses those issues that affect the design of the facilities included within the Project scope of work.

The Facilities to be covered under the project scope are:

- Satellite Manifold at BBOU complete with all Process, Utility & Control Infrastructure, however with provision for PIG receiver in phase 2 production from OPUG.
- Evacuation Manifold at ELEP complete with all Process, Utility & Control Infrastructure, however with provision for PIG receiver in phase 2 production from NUNR.
- Pipeline from BBOU to ELEP
- Pipeline from ELEP to AKAB (up to Inlet Flange of Slug Catcher)
- PIG Launchers at BBOU and ELEP Manifold
- PIG Receivers at ELEP and AKAB
- Manifold control system also to be provided. Control Room required at each Manifold to house the Manifold Control system & Telecom system
- Cabinet Room to house the electrical MCC.
- UPS Power for Control System, associated Civil, Mechanical, Pipeline, Electrical, Instrumentation & Control, Telecommunication, and all other supporting infrastructure.

The Scope of the FEED Engineering covers the provision of all necessary information, technical drawings, philosophies, specifications, design data, listing of codes and standards required for the detailed design and construction of the Multiphase pipeline system.

2. Regulations, Codes And Standards

The latest versions of the regulations, codes, and standards and extant amendment or supplements shall apply. Local laws regulating the oil and gas industry in Nigeria shall be strictly adhered to. The order of precedence for the documents shall be as follows:

- Nigerian Local Standards and Codes
- The Contract
- Project Specifications
- Employer's Standards
- Shell DEPs
- International Codes and Standards

In all cases, the most stringent shall apply except where constraints dictate otherwise.

Conflicts between specifications and referenced Codes and Standards or deviations thereof shall be referred in a Technical Query (TQ) to the EMPLOYER for resolution.

2.1. Shell Design and Engineering Practices (DEPs)

Applicable DEPs, supplements and MESC SPE Codes are as referenced in various discipline dossiers, some of which are listed below.

Document Number	Title
DEP 80.00.10.11	Gen Layout of Onshore Facilities
DEP 30.00.60.10-GEN	Human Factors Engineering in Projects
DEP 33.64.10.10	Electrical Design
DEP 80.45.10.10	Pressure Relief, Emergency Depressurising, Flare and Vent Systems
DEP 34.14.20.31	Drainage Systems and Primary Treatment Facilities
DEP 80.47.10.30-Gen	Assessment of Fire Safety of Onshore Installations
DEP 32.30.20.11	Fire, Gas and Smoke Detection
DEP 80.00.10.10	Area Classification (Amendments/Supplements to IP 15)
DEP 80.47.10.31-Gen:	Active Fire Protection Systems and Equipment for Onshore Facilities
DEP 80.47.10.32-Gen:	Portable and Mobile Equipment for Fire-Fighting

2.2. International Codes And Standard

Discipline	Document Number	Title
		Area Classification Code for Installations Handling Flammable Fluids, Institute of Petroleum Model Code of Safe Practice, Part 15, 3rd Edition
	IEC 61508	Functional Safety of Electrical/Electronic/Programmable Electronic Safety-related Systems
	IEC 60079	Explosive Atmospheres – Part 0: Equipment – General requirements

3. Hazard Management Philosophy

3.1. General

The hazard management philosophy for the project is that all credible and foreseeable hazards and their consequences will be identified, their effects assessed and measures engineered to reduce risks to both tolerable and ALARP.

These objectives will be achieved by adopting the following hazard management hierarchy (in order of preference):

- Elimination and minimisation of HSSE hazards by using options with a lower impact on HSSE;
- Substitution by using products and/or processes with a lower impact on HSSE;
- Isolation / separation of hazards and targets;
- Engineering controls – prevention;
- Engineering controls – mitigation;
- Organisational controls, i.e. competence and communication;
- Procedural controls;
- Personal Protective Equipment.

The design will use a risk based-approach to assess the potential consequences from hazards associated with the production facilities, handling and storage of hazardous materials.

3.2. Hazard & Effect Management Process (HEMP)

The Hazards and Effects Management Process (HEMP) addresses HSSE risks for the project in a systematic way and is the critical element within the HSSE-MS framework ensuring that HSSE hazards and their potential effects are fully evaluated. HEMP consists of varied processes each having the purpose to identify hazards, assess their potential effects, control the threats and recover from events, such that the HSSE related risks are reduced to a level which is As Low As Reasonably Practicable (ALARP).

HEMP activities that will be performed to demonstrate that all safety risks have been identified and risks reduced to a level that is ALARP are shown in Table 2 include:

Activity	Comments
Health, Safety, Security and Environment (HSSE) & Social Performance (SP) Management System (MS) and Activity Plan	Plan will identify all the deliverables that are to be produced.
Hazardous Area Classification Schedule	Required for the remote platforms as an input to the Hazardous Area classification drawings.
HSSE Hazards and Effects Register	Output from the HAZID and other hazard identification exercises
Design ALARP Demonstration Report (Design HSSE Case)	Report demonstrating that the risks have been identified and demonstrated to be tolerable and ALARP. This will be required for the pipelines and remote platforms.
Hazard and Operability (HAZOP) Report during FEED and Detailed Design	HAZOPs to be carried out on the Process Engineering Flow Schemes (PEFS)
HSSE Design Philosophy (This document)	
Hazard Identification (HAZID) Report	Required to identify the hazards to personnel, environment, asset and community.
Major Hazard Assessment	This will seek to quantify the MAHs and the mitigation measures against them.
Safety Critical Elements and Performance Standards Assessment/ Bowtie Review	
Fire, Gas Dispersion and Explosion Assessment	This will be carried out for the remote platforms to confirm that offsite risks are minimised.
Vent Dispersion / Radiation Study	Required to assess dimensions of vent stacks at the remote manifolds
Human Factors Engineering Screening Report	Screening to be carried out at early opportunity to set out Human Factors action plan for project.

3.2.1. Hazard Identification

The primary hazards associated with the project facilities are likely to be events caused by the release of hydrocarbon gas/condensate from the pipelines/piping and associated equipment. Due to the presence of significant quantities of hydrocarbon gas/condensate, the most significant incidents are considered to be explosions and jet fires. It is these incidents which have the highest potential for escalation and the most significant impact on the integrity of the installation and safety of personnel.

The following selection of applicable HEMP tools for the project have been identified as appropriate and effective in the identification and assessment of the hazards and risks of the facilities:

- HAZID
- Layout Review;
- HFE (Human Factors Engineering) Analysis;
- Hazard and Operability (HAZOP) Review;
- Fire and Explosion Risk Assessment;
- Safety Integrity Level Review;
- Bow Tie Analysis;
- Consequence Analysis

A Hazard and Operability Study (HAZOP) shall be undertaken using the Engineering Flow Schemes prior to issue of "Approved for Design" drawings. A further HAZOP shall be undertaken during detailed design to take account of any changes that may arise during the development of the design together with vendor information.

3.2.2. Hazard Analysis

During the FEED phase preliminary analysis of the main hazards identified will help to ascertain the level of risk associated with the facilities. Specific Fire and Explosion Risk Assessment Models will be developed for the remote platforms to provide input to project safety studies and to support any cost benefit decisions. Where appropriate, they will be used to demonstrate that risk targets have been met and that the risks to personnel are ALARP.

3.2.3. ALARP Criteria

The Design & Operations HSSE Cases demonstrate that all severity 5 and high-risk hazards have been identified, the risks from the hazards and effects evaluated and understood and that risk reduction measures to manage HSSE risks from these major hazards are in place, using bowtie methodology.

In assessing the risks associated with the facilities HSSE hazards, the level of risk assessment decision-making process is chosen based on the guidance chart shown below.

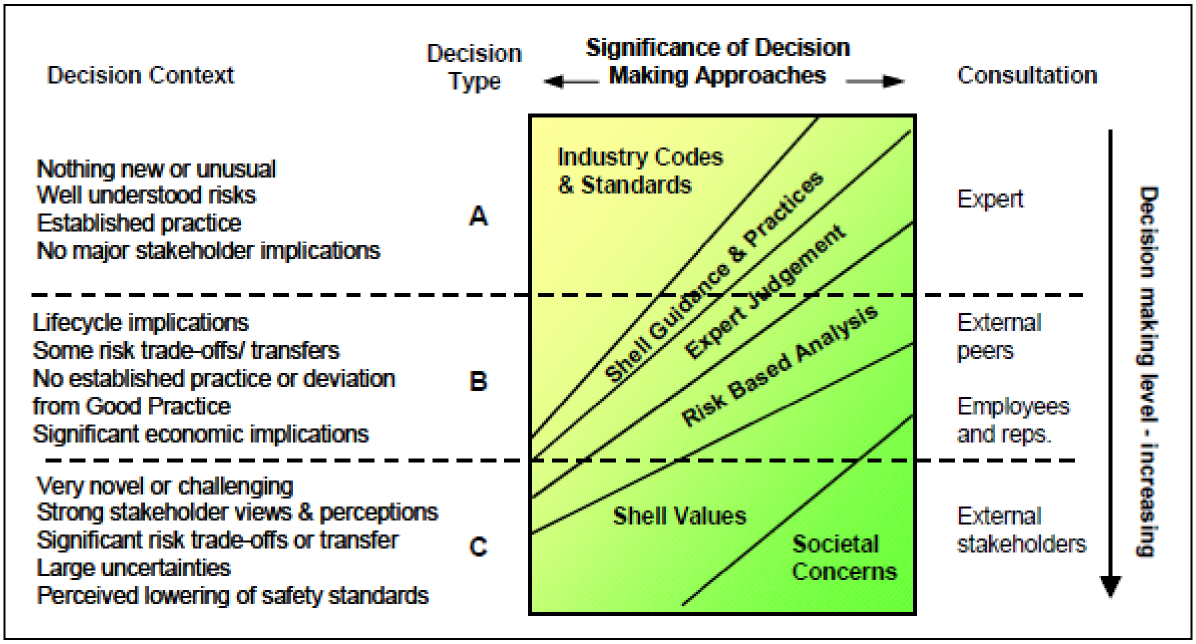

Figure 2: Decision Making Framework

A preliminary review of the hazards associated with this project facilities concluded that the hazards are well understood and the issues considered for risk reduction lie in Band A of the Framework whereby codes and standards, good practice, engineering judgment are the main factors in determining whether a measure is reasonably practicable. The only exceptions are the risk reduction issues against the consequences of fire/explosions which lie in Band B whereby some element of risk-based analysis dominates e.g. occupied building risk assessments.

The following philosophy to determine tolerability and ALARP will be adopted by the project, namely:

- The design is of low complexity and uses conventional oil and gas processing equipment. Accordingly, the hazards are well understood and there is nothing novel or unusual about the proposed design.
- The layout is very open with significant spacing between major pieces of equipment and no confined structures.
- Detailed technical safety studies will be conducted to demonstrate that all major accident hazards have been identified and risks reduced to a level that is ALARP.
- FRED and Shepherd Process safety Risk Tools shall be used to model the remote platforms to ensure that onsite and offsite risks are quantified and confirmed as being tolerable and reduced to ALARP.

3.3. Safety Critical Elements

Safety critical elements (SCE) shall be identified and performance standards and performance assurance tasks developed for the Design, Construction, and Commissioning and Operation phases. During the FEED phase, a complete list of all applicable Safety Critical Elements (SCE) and their associated Performance Standards (PS) shall be developed. The SCEs and PS shall be maintained through the projects' life cycle and shall be validated at the end of each of the project phases.

Consideration of safety-critical elements shall include systems for the prevention, detection, control and mitigation of major accidents. Items improving reliability by providing redundancy or diversity shall also be considered. The HEMP process requires that PS shall be developed for the Safety Critical Elements.

The SCEs shall be identified through the use of bowties which will assist with demonstrating that sufficient barriers are in place.

4. Fire Gas And Explosion Philosophy

The philosophy adopted for the project is that all credible potential flammable gas, fire and explosion scenarios are identified and measures engineered such that residual risks are ALARP.

Measures that will be implemented to reduce flammable gas, fire and explosion risks are described within Section 6 (Control measures) and Section 7 (Recovery Preparedness Measures).

These measures shall demonstrably provide the optimum type, quality and quantity of protection such that:

- The likelihood of loss of containment of a flammable fluid is minimised as far as is reasonably practicable.
- The inventory of flammable gas is minimised and the likelihood of personnel being exposed following a loss of containment, is minimised as far as is reasonably practicable.

And, in the event of a flammable gas release, fire or explosion:

- The intensity and duration of flames and intensity of explosion overpressures are minimised as far as is reasonably practicable.
- Two diverse means of escape shall be provided from all areas of the facility.
- Emergency systems maintain their integrity such that personnel are protected and are subsequently able to evacuate the site, and automatic control systems (e.g. isolation, shutdown, blowdown) are able to function.
- Escalation is prevented or minimised (hence assets protected and threat to people and environment minimised).

5. Control of Hazards

The optimum way of controlling hazards is to prevent hazardous events from occurring. The emphasis in design should therefore be on removing hazards altogether (inherent safety). If it proves impossible to remove the hazards completely, then efforts should be taken to reduce the probability of hazardous events occurring.

5.1. Principles of the Facilities Design to Minimise Risk

This section sets out the principles behind control techniques that will be adopted in the design of the facilities. Since hydrocarbons contained within process equipment and piping will present a significant hazard, the control techniques focus primarily on measures that will prevent or reduce the likelihood of fires and explosions.

5.1.1. Design for Ease of Operation and Maintenance

The overall process configuration should be as simple as possible for the functions the facilities have to perform. Project reviews during front end and detail design must ensure that the means of controlling the facilities is clear, provision for process isolation is adequate, operators have good access to all equipment, appropriate lifting devices are provided and provision has been made for non-routine operations.

5.1.2. Substitution of Hazardous Materials

Where possible the use of hazardous materials should be eliminated by either selection of alternative materials or adopting alternative processes.

5.1.3. Reduce Inventory of Toxic/ Hazardous Materials

Volumes of stored hazardous chemicals and process fluids should be minimised with fire-safe, actuated shutdown valves segregating significant process inventories.

5.1.4. Simplify the Process

Whenever possible the design should minimise high pressure/ low pressure interfaces and eliminate complex components.

5.1.5. Reduce Probability of Loss of Containment

Mechanical design should provide appropriate mechanical integrity, i.e. minimise flanges, small bore and screwed fittings, use effective seals, provide protection from mechanical damage by mechanical barriers, layout or shutdown. Providing adequate support and protection against mechanical failure as a result of vibration, corrosion or over-pressuring of blocked-in pipe sections is a fundamental part of process and piping design.

5.1.6. Eliminate Sources of Ignition

Potential ignition sources should be located outside of hazardous areas where possible. Where equipment is required to be located within hazardous areas then it must be rated for the appropriate zone as per IP Part 15

5.1.7. Accessibility to Equipment

The facilities shall be designed for adequate access and working space around equipment

5.1.8. Isolation Philosophy

Provision for proper isolation of process plant shall be made.

5.1.9. Human Factors Philosophy

The impact of Human Factors Engineering (HFE) on process safety shall be adequately considered. Proper integration of human factors guidelines, tools and techniques into the design of the facilities shall be achieved, which will lead to improved operator performance and has the potential to lead to a reduction of life cycle costs.

The critical tasks and activities shall be identified and assessed via an HFE Screening and application of appropriate HFE Design analysis tools.

Critical operational activities as well as specific emergency response philosophies and procedures are to be addressed from a human factors point of view with the objective to minimise inherent risk of human error.

5.2. Specific Controls Incorporated in Design to Minimise Risk

5.2.1. Inherent Safety

In line with inherent safety principles, the following measures shall be incorporated into the design:

- All process fluid piping joints shall be welded unless demonstrated as being either impracticable or unreasonable.
- Double block and bleed (DB&B) valves shall be used instead of individual valves, where pressure class requires DB&B (600# and above).
- All vents and bleed points from equipment that could potentially contain process fluid shall be hard piped into the vent header.

6. Recovery from Hazardous Events

This section focuses on the recovery measures that will be incorporated in the design to mitigate the consequences should a hazardous event occur. Recovery measures shall include provisions to:

- Detect any potential, or actual, loss of containment.
- Isolate the facility inventory.
- Minimise the duration of any event by reducing pressure and inventory.
- Provide appropriate separation between equipment to minimise escalation.
- Minimise the risk of escalation by reducing the probability of ignition.
- Reduce the effects of any resultant explosions.
- Ensure emergency power and communication facilities.
- Allow escape and evacuation of personnel.

6.1. Fire and Gas Detection

6.1.1. Design Philosophy

The early detection of flammable and toxic gases and/or fire is essential to prevent minor incidents escalating into major events. Automatic detection will be the primary means of identifying these types of release. Full details of the Fire and Gas Philosophy will be provided in the Fire and Gas Detection Design Philosophy.

The fire safety philosophy for the project is to achieve the desired degree of fire safety primarily by incorporating the control techniques described in Section 6. Where potential fire hazards still exist after implementation of these control measures, then fire and gas detection systems shall be provided.

Main hazards to be detected are as follows:

- Flammable gas clouds with sufficient size to cause an explosion.
- Flammable gas clouds migrating towards safe areas or ignition sources.
- Fires.

6.1.2. Detection Requirements

Where provided, F & G detection system has the following functional requirements:

- Provide continuous automatic monitoring for fire and hazardous accumulations of flammable/toxic gases.
- Monitor for ingress of smoke and flammable/toxic gas into areas where they may represent a hazard.

- Permit manual or automatic initiation of an alarm.
- Alert people that may be present on the platform at the time to the presence and location of a fire or hazardous accumulation of flammable gases. The manifolds are unmanned but will have a guard on duty continuously and regular visits by Operators from the main base in Akabel
- Minimise the number of spurious trips and alarms and contain appropriate facilities.
- Initiate an appropriate response action.
- Initiate an appropriate level of control action.
- Be capable of operating under the conditions experienced at the time F&G detection is needed.
- Use field devices suitable for the area in which they will be located.

The number and location of fire and gas detectors required shall be estimated during the FEED phase and be determined based on those areas that are considered higher risk and industry guidelines.

The exact numbers, types and locations of fire and gas detectors shall be subjected and confirmed by a Fire and Gas mapping study during detail design to achieve optimised coverage and response time. The Fire and Gas mapping study shall be performed by the contractor using Fire and Gas mapping software. Based on the Fire and Gas Mapping study, the Contractor shall optimize the fire and gas detectors layout.

6.1.3. Fire and Gas System Design

The type, selection and specification of fire, gas and smoke detectors shall follow the requirements in DEP 32.30.20.11

All detectors shall be interfaced with a fire and gas logic system which shall be linked to the emergency shutdown system. Fire and gas status and alarms shall be displayed on a fire and gas matrix panel with the Control Room and dedicated graphics shall be configured within the DCS to provide a complete overview of the system by fire zone.

Response measures in the form of alarms and executive actions shall accord with the following high-level guidelines:

- Alarms shall be annunciated via dedicated fire and gas matrix panel within the Control Room.
- A voting system shall be incorporated within the Fire and Gas system to reduce spurious alarms without unduly extending confirmed detection times.
- Personnel shall be alerted to the potential for imminent danger by audible and visual means, via the Public Address / General Alarm system and flashing beacons, so that they may escape from the area.

- Confirmed fire or gas detection shall initiate ESD and emergency depressurisation (EDP).

6.1.4. Detection by Persons

Personnel detecting loss of containment, smoke, fire, flammable or toxic gas release shall be given efficient means to alert other personnel or call for assistance to deal with the incident.

Means to alert others shall be via a personnel radio link with the control centre, manual alarm call points and direct telephone links.

6.1.5. Portable Gas Detection

Portable gas detection shall be provided for monitoring, for example in any low points, for example separator or drains tank pits.

6.1.6. Emergency Shutdown and Emergency Depressurisation

The project facilities shall be provided with an ESD system and blowdown, in line with DEP 80.45.10.10 Pressure Relief, Emergency Depressuring, Flare and Vent Systems

ESD and blowdown of pressurised vessels and piping is an acknowledged way to reduce the likelihood of escalation from accidental hydrocarbon release incidents. This is achieved by isolation of the main boundaries and by disposing of the hydrocarbon inventory in a safe and controlled manner and reducing the pressure. Both of these measures act to reduce the magnitude and duration of the hazardous event.

The intent of this section is to provide a high-level safety philosophy to guide the isolation and emergency depressurisation requirements of the plant during an emergency. Full details of the process safeguarding will be provided in the Control and Safeguarding Philosophy.

6.1.7. Emergency Shutdown Philosophy

The ESD system shall be designed to:

- Isolate the process facility from the hydrocarbon inventories in the pipelines entering and leaving the site. The main isolations shall be achieved using the pipeline ESD valves.
- Segregate significant inventories to limit the quantity of material released on loss of containment.
- Control potential ignition sources by shutdown of non-intrinsically safe equipment in hazardous areas.
- Initiate emergency depressurisation (blowdown).

The system shall be designed so that it is capable of fulfilling its function under the conditions which may be experienced when the system is required to operate.

ESD valves plus their actuating system shall remain operational for at least 15 minutes in the event of any fire or explosion, or for as long as necessary to fulfil their intended function, whichever is longer. Where practical, this shall be achieved by locating the valves outside the area where they could be impaired by fire or explosion. Should locating valves outside of the fire hazard area be impractical, passive fire protection shall be considered for the pipeline ESD valves, their actuators and associated components e.g. structural supports. Valves and actuators shall be arranged for fail safe operation.

Manual alarm call points and ESD pushbuttons shall be located at a safe distance from the fire risk area, preferably on exit routes.

6.1.8. Blowdown Philosophy

The principal objectives of blowdown will be:

- To mitigate the consequences of a hydrocarbon release by reducing the leakage rate and system inventory; and
- To avoid catastrophic rupture of the separators in a fire scenario, by reducing the internal pressure to lower the stress.

For ignited hydrocarbon releases, blowdown is a major risk mitigation factor because the decay of system pressure results in shorter jet flames and lessens the duration of the release. This significantly reduces the escalation potential by reducing the probability of flame impingement onto other sections of process plant or key structures. In addition, by reducing the pressure, the stresses on a vessel under fire attack will reduce, will be less likely to reach dangerous levels and thus the probability of catastrophic vessel failure will be diminished.

The intent is that a blowdown system should have adequate capacity to permit reduction of vessel stress to a level at which stress rupture is not of immediate concern.

The blowdown strategy for the project shall be to depressurise as rapidly as possible within the constraints of the flare design at Akabel facility, liquid knockout and restrictions on individual equipment (to avoid low temperature hazards).

6.2. Facilities Layout

6.2.1. Layout Philosophy

The philosophy for facilities layout shall follow the guiding principles in the; DEP80.00.10.11-Gen Layout of Onshore Facilities

The layout of the facilities shall be developed to ensure that risk levels are reduced to ALARP and to optimize operability and maintainability. Important objectives of layout development include:

- Maximise separation between low-risk areas (control room, utilities, etc.), and higher risk areas (high pressure hydrocarbon containing equipment).
- Adequate provision for operations and maintenance.
- Locate equipment to minimise the risk of loss of containment and escalation.
- Minimise the risk of ignition of a flammable release.

- Minimise the overpressure from a delayed ignition of a flammable release.
- Provide suitable escape routes from the process areas to muster areas and proper access for emergency response.
- Allow space for future expansion, including construction and storage areas.

These objectives will be achieved for the project by adopting the following strategies:

- Maximise practical physical separation between the process equipment and buildings/ignition sources, i.e. e\weather shelter.
- Locate process equipment in an open area and optimise process equipment and piping layout to minimise congestion, thereby minimising the potential for gas build-up and blast overpressure.
- Locate guard building upwind of the process area, with respect to prevailing wind direction, to minimise the likelihood of a flammable gas cloud reaching the safe area and to optimise escape from the hazardous areas.
- Locate vents crosswind of process area, with respect to prevailing wind direction.
- Optimise layout with respect to escape routes. People should be able to escape away (from two opposite escape routes) from the hazardous areas within the platform. Availability of at least one escape route under all credible events shall be demonstrated.
- Consider requirements for future equipment, including space for future construction activities and laydown area.

6.2.2. Minimum Separation Distances

As required by the HEMP, the minimum separation distances within the facility shall be confirmed based on the results of consequence modelling performed for credible hazardous events identified as part of the HAZID and/or Fire and Explosion Risk Assessment modelling.

Layout and distances between equipment shall be such that in the event of a gas release, fire or explosion, the probability of escalation is reduced to ALARP.

6.2.3. Manned Buildings

There are no manned buildings. As part of the design for the platforms, weather shelters is provided at each of the remote platforms. These shelters should be designed such that the risks to personnel within the houses are reduced to ALARP. The design of the shelters in relation to control and mitigation against accidental events will be assessed in fire, gas dispersion and explosion assessment

6.2.4. Drainage

The philosophy for open and closed drainage arrangements for the facilities will be set out in the Drainage Philosophy. Key objectives can be summarised as follows:

- Minimise the probability of creating large pools of fuel under or in the vicinity of equipment containing flammable liquids or vapours.
- Allow for safe maintenance of facilities (including vessel inspections) by draining process fluids and recovering them to the process for disposal.
- Minimise oil contamination of rainwater.
- Contain minor spills within bunded areas.
- No open connections to the closed drain system; such as liquid traps, tundishes or surface water drains.

The design of the drain systems shall follow the requirements in DEP 34.14.20.31

6.3. Fire Protection

6.3.1. Fire Safety Philosophy

The fire safety philosophy for the project facilities is to ensure that risks due to fire hazards are ALARP. This will be achieved primarily by prevention controls. Where, after preventative measures have been applied, and potential fire hazards still exist, fire protection systems shall be provided to limit or prevent fire escalation to avoid risk to life and minimise material damage. The intent of this section is to establish the basis on which the appropriate level of fire protection will be determined and to provide high level guidance on the fire protection measures.

6.3.2. Active Fire Protection

6.3.2.1. Active Fire Protection Philosophy

No firewater systems are to be installed for the project facilities. Reliance will be placed on the isolation and blowdown of hydrocarbon inventories to minimise the consequences of a fire.

- There is minimal (if any) pool fire potential on the site as inventory is primarily gas and light condensate (which will rapidly evaporate at ambient conditions) and there is no storage on site;
- The blowdown philosophy intends to reduce the onsite pressure to at least 6.9barg within 15 minutes thereby minimising the duration of any jet fires and associated escalation risk;
- Fire and Gas detection will be provided to facilitate early blowdown when required
- The site layout philosophy and early draft layouts indicate that the requisite distances between equipment can be met thereby further reducing the escalation potential;
- Asking site personnel to fight large scale fires with monitors puts them at a higher risk than simply allowing the system to depressurise thereby removing the inventory.

6.3.2.2. Electrical Equipment Room / Remote Terminal Unit

Rooms with electrical/electronic control equipment where provided shall each have two carbon dioxide extinguishers located preferably inside the room near each entrance. Additional extinguishers may be required inside the room as well if the travel distance exceeds 6m.

6.3.2.3. Weather Shelters

The travel distance to extinguishers located in weather shelters shall not exceed 30m. Depending on the fire hazard, a carbon dioxide extinguisher or a foam spray extinguisher shall be selected.

6.3.2.4. Passive Fire Protection

The objective of passive fire protection (PFP) of equipment and steel structures is to prevent the escalation of fires to an unacceptable level by providing temporary protection until full fire-fighting capabilities can be deployed.

PFP shall be considered for the project facilities where it is essential that equipment or system integrity be maintained during a fire. Fire scenarios and the potential for escalation will be evaluated and areas requiring PFP will be identified taking account of other recovery systems available (e.g. blowdown, active fire protection).

To minimise the occurrence of under-lagging corrosion and to avoid additional cost, passive fire protection will only be installed where the fire hazard assessment justifies the need.

6.4. Explosion Mitigation

The project facilities shall be located in the open and process equipment and piping layout shall be optimised to maximise physical separation and minimise congestion. Accordingly, the potential explosion overpressures and explosion risks should be low. Protection of the weather shelters from explosion overpressure shall be achieved by physical separation and appropriate construction.

6.5. Ignition Source Control

6.5.1. Hazardous Area Classification

Hazardous areas will be determined based on DEP 80.00.10.10 which is based on the Area Classification Code for Installations Handling Flammable Fluids, Institute of Petroleum Model Code of Safe Practice, Part 15, 3rd Edition

Continuous grades of release giving rise to a Zone 0 area shall be confined to essential vents that cannot be flared – these shall be limited to exceptional cases justified on a risk basis in accordance with the no-venting policy.

Hazardous area schedule and hazardous area classification drawings showing the plan view and elevations shall be prepared and include the following information:

- Identification of sources of release.
- All ventilation inlets and outlets.
- Air intakes and exhausts of all internal combustion machinery.
- Location of all equipment units.
- Points of air transfer from modules which may affect classification.

- Tank or process vents.
- The classification and extent of all hazardous zones.
- Ventilation type.

A hazardous area schedule shall be produced identifying potential sources of emission, the process material, its operating conditions and flash point, fluid category, hazardous area boundary dimensions from source, source of release grading.

The weather shelter shall be in a non-hazardous area.

6.5.2. Electrical Equipment

Electrical equipment located in hazardous areas shall be selected and installed in accordance with DEP 33.64.10.10 and IEC 60079

Any electrical field equipment that is required to operate under emergency conditions shall be suitable for use within a Zone 1 hazardous area.

6.5.3. Non-Electrical Ignition Sources

All mechanical equipment installed in hazardous areas shall be manufactured to ensure that rotating parts are non-sparking and adequately protected against the generation of a static charge. Surface temperatures shall not exceed 200°C. The use of aluminium shall be restricted.

6.6. Escape and Evacuation

Escape routes shall be provided on the remote platforms to enable personnel to reach the designated muster area in an emergency. The following principles shall be adopted in the design of escape routes:

- Two diverse means of escape shall be provided from all process areas.
- Sufficient emergency exit gates shall be provided to enable personnel to readily evacuate from the site in all Major Accident Events.
- The minimum width of escape routes shall be 1m and a headroom of 2.1m shall be provided.
- All escape routes must permit the transfer of personnel who may be injured including those who may need to be moved by stretcher
- Escape routes and emergency exit gates shall be marked and lit so they are readily identifiable by personnel in an emergency.
- A designated muster area shall be provided outside of the fence line.
- Emergency lighting shall provide minimum lighting for a period of 60 minutes to enable controlled movement of personnel should the main power supply fail.

7. Health

7.1. Chemicals

The project will manage the risk of exposure to chemicals by following the requirements of the Shell HSSE & SP Control Framework Acute Toxic Substances requirements. A Health Risk Assessment for the project will be carried out during FEED stage and control and mitigation measure will be specified accordingly.

7.2. Noise

Noise levels from equipment in the project facilities shall be as far as reasonably practicable. This shall be primarily achieved by reduction at source, i.e. equipment which is a potential noise source should be requested to be low noise emitting.

Nigerian legislative limits shall be complied with as presented within the Petroleum Act (CAP 350 LFN) Mineral Oils (Safety) Regulations 1997 which states the following:

The manager shall ensure that:

- Workers are provided with the appropriate hearing protection if noise levels are equal to or greater than 85 dBA for an 8 hour time weighted average (TWA).
- No person shall, unless appropriately protected, be exposed to noise level equal to 115 ~A or greater for any length of time, notwithstanding that the TWA is below 85dBA action level.
- The sound pressure level at the edge of the nearest residential area shall not exceed 50dBA at night.

The equivalent noise intensity shall not exceed 90dBA during the daily work shift (8 hours)

8. Environment

The project facilities will be designed to comply with Nigerian regulatory requirements.

The project facilities shall be designed and operate to ISO-14001 certification standards. This section defines the key parameters that form the environmental basis of design. The aim is to demonstrate that environmental risks are tolerable and impacts reduced to as low as reasonably practicable (ALARP). The over-arching philosophies for managing impacts to the environment include

- Energy use, energy optimisation and minimisation of atmospheric emissions shall be evaluated as part of the decision making process when assessing options for the main energy consumption items.
- Key design decisions regarding venting shall aim to minimise atmospheric emissions associated with the project facilities
- Production of non-gaseous waste shall be avoided, where possible, by using clean technologies or using waste minimisation/waste recovery technologies
- Generation of wastes and discharges of wastewater shall be minimised.

The environmental basis of design shall be reviewed and updated as the project matures to reflect any design decisions that might have an impact on the environmental performance. Relevant findings of future lessons learned reviews, as well as recommendations received through the update of the Environmental Impact Assessment report, shall also be taken into account.

8.1. Atmospheric Emissions

A Greenhouse Gas (GHG) and Energy Efficiency Plan shall be prepared during FEED, to identify, quantify potential GHG emissions and propose project opportunities to improve energy efficiency. The GHG & EE plan will propose measures to ensure atmospheric emissions are minimized. Energy use and equipment selection should be optimised for normal and maximum operations and should take into account the full operational lifecycle of the facility including turndown/late life. Emissions of SOx and NOx from point sources in all Project installations should comply with local legal requirements.

8.2. Venting

There shall be no continuous venting of hydrocarbons. A Greenhouse Gas (GHG) and Energy Efficiency Plan shall be developed during FEED to identify, quantify potential GHG emissions and propose project opportunities to improve energy efficiency. Further project opportunities to reduce venting shall be identified. Design shall incorporate minimisation of potential leak sources to avoid fugitive emission and provision of a closed drainage system for the hydrocarbon systems, operational controls shall be established and maintained to ensure VOC emissions are reduced to As Low As Reasonably Practicable

8.3. Chemicals

Design shall:

- Minimise the use of chemicals
- Not use chemicals with a substitution warning and
- Ensure an auditable chemical assessment and selection process is used to demonstrate chemicals with the least environmental impact are adopted, where practicable.
- Ensure fit for purpose bunding is provided around tanks containing chemicals, lube oils, diesel, etc.

8.4. Spill Prevention

To minimise the potential for and to reduce risks of leaks and spills at source, the design shall ensure that:

- Leak paths are minimised.
- Bunded areas are designed fit for purpose and adequate space allowance is provided.
- Diesel tanks and re-fuelling location requirements are assessed.

8.5. Waste

The design shall ensure sufficient space allowance for Waste containers to allow adequate segregation and storage of Waste on the sites. Design shall: Incorporate controls to reduce Waste generation into Procedures and working practices. Where appropriate: - Identify opportunities to reuse Waste for the same or alternative applications, including in other industries, or allow for provision to return unused materials to suppliers. - Identify recycling and recovery opportunities for Waste material.

8.6. Impact Assessment

There shall be Project EIA report, subjected to the EIA iterative process and approvals. Several formal engagements shall be undertaken with DPR, Federal Ministry of Environment (FMEnv) regarding impact assessment scoping and supporting studies to ensure that complete update of the EIA early in FEED phase and subsequent update in the later phases of the project. The Environmental Impact Assessment (EIA) to be completed under the Nigerian EGASPIN Environmental Regulations and in accordance with all other applicable Nigerian and international legislation

9. Security

A security risk assessment shall be carried out during FEED and adequate measures shall be provided to mitigate security risks to tolerable level and ALARP using both physical and procedural security system requirements